

STEINKJER, Norway

WMO: 012770

Lat: 64.022N

Lon: 11.451E

Elev: 262

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.2	8.4	-1.4	5.1	5.9	2.3	6.3	10.7	18.8	37.8	16.5	36.7	3.9	90	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	78.7	63.4	75.1	61.6	71.8	60.3	65.3	75.8	63.2	72.9	61.3	70.0	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	80.6	70.4	58.9	75.0	67.9	57.2	70.4	65.4	30.3	76.2	28.8	72.7	27.4	70.0	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.1	11.2	0.3	84.5	4.7	3.8	-3.0	87.3	-5.7	89.5	-8.4	91.7	-11.8	94.5		
(5)			-0.2	69.1	4.6	2.2	-3.6	70.7	-6.3	72.0	-8.8	73.2	-12.2	74.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	27.2	27.9	32.1	40.5	48.8	54.5	60.2	58.7	51.7	41.8	33.8	30.0		
	DBStd	13.57	8.48	8.58	7.27	5.62	7.11	6.29	6.02	5.55	5.17	5.96	7.22	8.39		
	HDD50	3716	706	618	556	293	108	22	0	3	40	263	486	621		
	HDD65	8307	1171	1038	1021	736	502	318	175	206	398	720	936	1086		
	CDD50	928	0	0	0	7	72	159	316	274	92	8	0	0		
	CDD65	45	0	0	0	0	2	5	26	12	0	0	0	0		
	CDH74	422	0	0	0	0	20	71	236	95	0	0	0	0		
	CDH80	65	0	0	0	0	1	11	47	6	0	0	0	0		
	WSAvg	4.7	4.9	4.9	5.5	5.1	4.9	4.8	4.1	3.9	4.2	4.3	4.4	4.9		
	PrecAvg	50.2	5.3	4.7	4.5	3.0	2.4	3.0	3.1	3.3	4.8	5.3	4.9	5.8		
(15) Precipitation	PrecMax	65.3	11.9	12.0	10.5	6.0	3.8	5.9	6.9	5.6	12.3	8.8	10.1	10.9		
	PrecMin	32.8	0.4	0.9	1.5	0.8	0.9	0.4	1.0	0.7	1.1	1.7	0.7	2.1		
	PrecStd	7.2	2.9	2.5	2.4	1.4	0.7	1.3	1.3	1.4	2.8	1.9	2.8	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	45.7	50.4	66.3	76.5	81.5	85.4	80.5	70.5	61.0	50.7	47.0		
		MCWB	40.1	40.0	41.6	50.5	58.4	63.1	66.4	63.2	59.3	52.4	47.4	42.8		
	2%	DB	41.4	43.2	46.1	59.5	71.6	74.9	80.1	76.6	65.8	56.5	47.6	44.3		
		MCWB	38.6	39.0	40.3	47.8	56.7	59.8	64.2	63.5	57.0	50.7	44.5	40.7		
	5%	DB	39.3	40.7	43.6	54.6	66.8	70.4	76.1	73.0	63.0	52.9	45.0	42.0		
		MCWB	36.6	37.3	39.1	45.1	54.4	57.9	63.0	61.7	55.3	48.6	41.9	39.1		
	10%	DB	37.0	38.4	41.6	50.1	61.4	66.6	72.4	69.4	60.0	50.0	43.1	39.8		
		MCWB	34.7	35.5	37.8	42.5	51.5	56.5	61.5	60.3	53.9	46.7	40.5	37.3		
	0.4%	WB	41.4	41.7	43.9	52.0	60.9	64.7	68.6	67.2	60.9	54.4	47.8	44.1		
		MCDB	42.9	44.6	47.4	64.2	73.3	76.7	79.7	77.5	68.0	58.6	49.3	45.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.4	39.8	41.4	48.7	57.6	61.9	66.2	64.6	58.2	51.1	45.0	41.4		
		MCDB	40.9	42.4	44.6	58.3	68.7	72.9	77.9	73.6	64.1	55.8	46.9	43.5		
	5%	WB	36.8	37.8	39.9	45.8	55.1	59.5	64.0	62.8	56.4	49.3	42.6	39.3		
		MCDB	38.6	40.1	43.2	53.1	66.1	68.3	74.3	71.1	61.3	52.4	44.6	41.5		
	10%	WB	35.1	35.8	38.2	43.5	52.2	56.9	61.8	60.8	54.7	47.1	40.5	37.6		
		MCDB	36.7	38.2	41.3	49.0	61.0	65.4	71.1	68.1	59.1	49.7	42.7	39.5		
	5% DB	MDBR	8.6	10.0	12.9	14.8	16.2	16.0	16.8	15.8	13.1	10.8	8.7	8.9		
		MCDBR	9.0	10.6	13.5	22.3	24.8	24.9	25.0	22.1	17.7	13.7	10.2	10.2		
		MCWBR	7.2	7.8	9.5	13.2	13.1	13.4	12.5	11.3	10.4	9.1	8.0	8.1		
		MCWBR	8.2	9.7	12.5	19.4	23.5	23.0	23.0	19.9	16.3	12.0	9.9	9.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.9	7.8	9.3	12.0	13.0	13.2	12.6	11.2	10.5	8.6	8.2	7.9		
	Clear-Sky Solar Irradiance	taub	0.227	0.291	0.322	0.347	0.348	0.348	0.355	0.343	0.327	0.304	0.353	0.359		
		taud	2.012	2.216	2.290	2.373	2.441	2.472	2.472	2.499	2.533	2.426	2.621	2.585		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	108	197	242	264	274	276	270	264	245	197	104	51		
		Edh at Noon	9	18	26	30	31	30	30	26	21	15	8	5		
(44) All-Sky Solar Radiation	RadStd		41	192	571	1119	1445	1530	1466	1125	656	287	69	18		
			6	31	93	98	96	175	150	124	74	41	12	2		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A	

Nomenclature: See separate page